



Hướng dẫn CRUD với JPA3.0 và Jakarta 6.0

1. Tạo entity

`@Entity`

`@AllArgsConstructor`

`@NoArgsConstructor`

`@Data`

`@Table(name = "categories")`

`@NamedQuery(name = "Category.findAll", query = "SELECT c FROM Category c")`

`public class Category implements Serializable {`

`private static final long serialVersionUID = 1L;`

www

```
@GeneratedValue(strategy = GenerationType.IDENTITY)

@Column(name = "CategoryId")

private int categoryid;

@Column(name = "CategoryName", columnDefinition = "nvarchar(50) not null")

@NotEmpty(message = "Không được phép rỗng")

private String categoryname;

@Column(name = "Images", columnDefinition = "nvarchar(500) null")

private String images;

private int status;

// bi-directional many-to-one association to Video

@OneToMany(mappedBy = "category")

private List<Video> videos;

public Video addVideo(Video video) {

    getVideos().add(video);

    video.setCategory(this);

    return video;

}

public Video removeVideo(Video video) {
```

```
video.setCategory(null);
```

```
return video;
```

```
}
```

```
}
```

```
=====
```

```
@NoArgsConstructor
```

```
@AllArgsConstructor
```

```
@Data
```

```
@Entity
```

```
@Table(name="Videos")
```

```
@NamedQuery(name="Video.findAll", query="SELECT v FROM Video v")
```

```
public class Video implements Serializable{
```

```
private static final long serialVersionUID = 1L;
```

```
@Id
```

```
@Column(name="VideoId")
```

```
private String videoId;
```

```
private int active;
```

```
@Column(name="Description", columnDefinition = "nvarchar(500) null")
```

```
private String description;
```

```
@Column(name="Poster" , columnDefinition = "nvarchar(500) null")
```

```
private String poster;
```

```
@Column(name="Title" , columnDefinition = "nvarchar(500) null")
```

```
private String title;
```

```
@Column(name="Views")
```

```
private int views;
```

```
//bi-directional many-to-one association to Category
```

```
@ManyToOne
```

```
@JoinColumn(name="CategoryId")
```

```
private Category category;
```

```
}
```

2. Viết các phương thức tầng Dao

```
public class CategoryDao implements ICategoryDao {
```

```
@Override
```

```
public void insert(Category category) {

    EntityManager enma = JPAConfig.getEntityManager();

    EntityTransaction trans = enma.getTransaction();

    try {

        trans.begin();

        enma.persist(category); //insert vào bảng

        trans.commit();

    } catch (Exception e) {

        e.printStackTrace();

        trans.rollback();

        throw e;

    } finally {

        enma.close();

    }

}

@Override

public void update(Category category) {

    EntityManager enma = JPAConfig.getEntityManager();

    EntityTransaction trans = enma.getTransaction();

    try {
```

```
enma.merge(category);//update vào bảng

trans.commit();

} catch (Exception e) {

e.printStackTrace();

trans.rollback();

throw e;

}finally {

enma.close();

}

}

@Override

public void delete(int cateid) throws Exception {

EntityManager enma = JPAConfig.getEntityManager();

EntityTransaction trans = enma.getTransaction();

try {

trans.begin();

Category category = enma.find(Category.class,cateid);

if(category != null) {

enma.remove(category);
```

```
    } else {

        throw new Exception("Không tìm thấy");

    }

    trans.commit();

} catch (Exception e) {

    e.printStackTrace();

    trans.rollback();

    throw e;

}finally {

    enma.close();

}

}

@Override

public Category findById(int cateid) {

    EntityManager enma = JPAConfig.getEntityManager();

    Category category = enma.find(Category.class, cateid);

    return category;

}

@Override
```

```
EntityManager enma = JPAConfig.getEntityManager();
```

```
String jpql = "SELECT c FROM Category c WHERE c.categoryname =:catename";
```

```
try {
```

```
TypedQuery<Category> query= enma.createQuery(jpql, Category.class);
```

```
query.setParameter("catename", name);
```

```
Category category= query.getSingleResult();
```

```
if(category==null) {
```

```
throw new Exception("Category Name đã tồn tại");
```

```
}
```

```
return category;
```

```
} finally {
```

```
enma.close();
```

```
}
```

```
}
```

```
@Override
```

```
public List<Category> findAll() {
```



```

EntityManager enma = JPAConfig.getEntityManager();

TypedQuery<Category> query= enma.createNamedQuery("Category.findAll", Category.c

return query.getResultList();

}

@Override

public List<Category> searchByName(String catname) {

EntityManager enma = JPAConfig.getEntityManager();

String jpql = "SELECT c FROM Category c WHERE c.catename like :catname";

TypedQuery<Category> query= enma.createQuery(jpql, Category.class);

query.setParameter("catename", "%" + catname + "%");

return query.getResultList();

}

@Override

public List<Category> findAll(int page, int pagesize) {

EntityManager enma = JPAConfig.getEntityManager();

TypedQuery<Category> query= enma.createNamedQuery("Category.findAll", Category.c

query.setFirstResult(page*pagesize);

query.setMaxResults(pagesize);

return query.getResultList();

```

@Override

```
public int count() {  
  
    EntityManager enma = JPAConfig.getEntityManager();  
  
    String jpql = "SELECT count(c) FROM Category c";  
  
    Query query = enma.createQuery(jpql);  
  
    return ((Long)query.getSingleResult()).intValue();  
  
}
```

=====

pull up thành interface

```
public interface ICategoryDao {  
  
    void insert(Category category);  
  
    int count();  
  
    List<Category> findAll(int page, int pagesize);  
  
    List<Category> searchByName(String catname);  
  
    List<Category> findAll();  
  
    Category findById(int cateid);
```

```
void delete(int cateid) throws Exception;
```

```
void update(Category category);
```

```
Category findByCategoryname(String name) throws Exception;
```

```
}
```

3. Viết các phương thức tầng Services

```
public interface ICategoryService {
```

```
void insert(Category category);
```

```
int count();
```

```
List<Category> findAll(int page, int pagesize);
```

```
List<Category> searchByName(String catname);
```

```
List<Category> findAll();
```

```
Category findById(int cateid);
```

```
void delete(int cateid) throws Exception;
```

```
void update(Category category);
```

```
Category findByCategoryname(String name);
```

```
}
```

```
public class CategoryServiceImpl implements ICategoryService {
```

```
    public ICategoryDao cateDao = new CategoryDao();
```

```
    @Override
```

```
    public List<Category> findAll() {
```

```
        return cateDao.findAll();
```

```
    }
```

```
    @Override
```

```
    public Category findById(int id) {
```

```
        return cateDao.findById(id);
```

```
    }
```

```
    @Override
```

```
    public List<Category> searchByName(String keyword) {
```

```
        return cateDao.searchByName(keyword);
```

@Override

```
public void insert(Category category) {  
  
    Category cate = this.findByCategoryname(category.getCategoryname());  
  
    if(cate==null) {  
  
        cateDao.insert(category);  
  
    }  
  
}
```

@Override

```
public void update(Category category) {  
  
    Category cate = this.findById(category.getCategoryid());  
  
    if(cate!=null) {  
  
        cateDao.update(category);  
  
    }  
  
}
```

@Override

```
public void delete(int id){
```

```
cateDao.delete(id);  
  
} catch (Exception e) {
```

```
    e.printStackTrace();  
}
```

```
}
```

```
@Override
```

```
public int count() {
```

```
    return cateDao.count();
```

```
}
```

```
@Override
```

```
public List<Category> findAll(int page, int pagesize) {
```

```
    return cateDao.findAll(page, pagesize);
```

```
}
```

```

public Category findByCategoryname(String name) {

    try {

        return cateDao.findByCategoryname(name);

    } catch (Exception e) {

        e.printStackTrace();

    }

    return null;

}
}

```

5. Viết Controller

```

@MultipartConfig()

@WebServlet(urlPatterns = { "/admin/categories", "/admin/category/add", "/admin/c

"/admin/category/edit", "/admin/category/update", "/admin/category/delete" })

public class CategoryController extends HttpServlet {

    private static final long serialVersionUID = 1L;

```

```

public CategoryService cateService = new CategoryServiceImpl();

@Override

protected void doGet(HttpServletRequest req, HttpServletResponse resp) throws Se

String url = req.getRequestURI();

if (url.contains("/admin/categories")) {

List<Category> list = cateService.findAll();

req.setAttribute("listcate", list);

req.getRequestDispatcher("/views/admin/category-list.jsp").forward(req, resp);

} else if (url.contains("/admin/category/add")) {

req.getRequestDispatcher("/views/admin/category-add.jsp").forward(req, resp);

} else if (url.contains("/admin/category/edit")) {

int id = Integer.parseInt(req.getParameter("id"));

Category category = cateService.findById(id);

req.setAttribute("cate", category);

req.getRequestDispatcher("/views/admin/category-edit.jsp").forward(req, resp);

} else {

int id = Integer.parseInt(req.getParameter("id"));

try {

cateService.delete(id);

} catch (Exception e) {

```



```
// TODO Auto-generated catch block
```

```
e.printStackTrace();
```

```
}
```

```
// chuyển trang
```

```
resp.sendRedirect(req.getContextPath() + "/admin/categories");
```

```
}
```

```
}
```

```
@Override
```

```
protected void doPost(HttpServletRequest req, HttpServletResponse resp) throws S
```

```
String url = req.getRequestURI();
```

```
if (url.contains("/admin/category/insert")) {
```

```
// lấy dữ liệu từ form
```

```
String categoryname = req.getParameter("categoryname");
```

```
int status = Integer.parseInt(req.getParameter("status"));
```

```
String images = req.getParameter("images");
```

```
// đưa dữ liệu vào model
```

```
Category category = new Category();
```

```
category.setCategoryname(categoryname);
```

```
category.setStatus(status);
```

```
String uploadPath = constants.DIR; // upload vào thư mục bất kỳ

File uploadDir = new File(uploadPath);

if (!uploadDir.exists())

uploadDir.mkdir();

try {

Part part = req.getPart("images1");

if (part.getSize() > 0) {

String filename = Paths.get(part.getSubmittedFileName()).getFileName().toString(

int index = filename.lastIndexOf(".");

String ext = filename.substring(index + 1);

fname = System.currentTimeMillis() + "." + ext;

part.write(uploadPath + "/" + fname);

category.setImages(fname);

} else if (images != null) {

category.setImages(images);

} else {

category.setImages("avatar.png");

}
```

```
    } catch (FileNotFoundException fne) {

fne.printStackTrace();

    }

    // đưa model vào phương thức insert

    cateService.insert(category);

    // chuyển trang

    resp.sendRedirect(req.getContextPath() + "/admin/categories");

    }

    if (url.contains("/admin/category/update")) {

    // lấy dữ liệu từ form

    int categoryid = Integer.parseInt(req.getParameter("categoryid"));

    String categoryname = req.getParameter("categoryname");

    int status = Integer.parseInt(req.getParameter("status"));

    String images = req.getParameter("images");

    // đưa dữ liệu vào model

    Category category = cateService.findById(categoryid);

    String fileold = category.getImages();

    category.setCategoryname(categoryname);

    category.setStatus(status);
```

```
String fname = "",
```

```
String uploadPath = constants.DIR; // upload vào thư mục bất kỳ
```

```
File uploadDir = new File(uploadPath);
```

```
if (!uploadDir.exists())
```

```
uploadDir.mkdir();
```

```
try {
```

```
Part part = req.getPart("images1");
```

```
if (part.getSize() > 0) {
```

```
// xóa file cũ trên thư mục
```

```
if (!category.getImages().substring(0, 5).equals("https") ) {
```

```
deleteFile(uploadPath+ "\\\" + fileold);
```

```
}
```

```
String filename = Paths.get(part.getSubmittedFileName()).getFileName().toString()
```

```
int index = filename.lastIndexOf(".");
```

```
String ext = filename.substring(index + 1);
```

```
fname = System.currentTimeMillis() + "." + ext;
```

```
part.write(uploadPath + "/" + fname);
```

```
category.setImages(fname);
```

```
    } else if (images != null) {

        category.setImages(images);

    } else {

        category.setImages(fileold);

    }

} catch (FileNotFoundException fne) {

    fne.printStackTrace();

}

// đưa model vào phương thức insert

cateService.update(category);

// chuyển trang

resp.sendRedirect(req.getContextPath() + "/admin/categories");

}

}

public static void deleteFile(String filePath) throws IOException {

    Path path = Paths.get(filePath);

    Files.delete(path);

}

}
```

category-list.jsp

```
<a href="<c:url value="/admin/category/add"/>">Add Category</a><br>
```

```
<hr>
```

```
<table border="1" width="100%">
```

```
<tr>
```

```
<th>STT</th>
```

```
<th>Images</th>
```

```
<th>Category name</th>
```

```
<th>Status</th>
```

```
<th>Action</th>
```

```
</tr>
```

```
<c:forEach items="${listcate}" var="cate" varStatus="STT">
```

```
<tr>
```

```
<td>${STT.index+1 }</td>
```

```
<c:if test="${cate.images.substring(0,5)=='https'}">
```

```
<c:url value="${cate.images }" var="imgUrl"></c:url>
```

```
</c:if>
```

```
<c:if test="${cate.images.substring(0,5)!='https'}">
```

```
<c:url value="/image/?name=${cate.images }" var="imgUrl"/></c:url>
```

```
</c:if>
```

```
<td></td>
```

```
<td>${cate.categoryname }</td>
```

```
<td>
```

```
<c:if test="${cate.status==1 }">
```

Hoạt động

```
</c:if>
```

```
<c:if test="${cate.status!=1 }">
```

Khóa

```
</c:if>
```

```
</td>
```

```
<td><a href="<c:url value='/admin/category/edit?id=${cate.categoryid }' />">Sửa</a>
```

```
| <a href="<c:url value='/admin/category/delete?id=${cate.categoryid }' />">Xóa</a>
```

```
</td>
```

```
</tr>
```

```
</c:forEach>
```

```
</table>
```

```
=====
```

category_add.jsp

```
<form action="<c:url value="/admin/category/insert"/>" method="post" enctype="mul

<label for="fname">Category name:</label><br>

<input type="text" id="categoryname" name="categoryname"><br>

<label for="lname">Link images:</label><br>

<input type="text" id="images" name="images"><br>

<label for="lname">Upload images:</label><br>

<input type="file" id="images1" name="images1"><br>

<label for="html">Status</label><br>

<input type="radio" id="ston" name="status" value="1">

<label for="css">Hoạt động</label><br>

<input type="radio" id="stoff" name="status" value="0">

<label for="javascript">Khóa</label>

<br><br>

<input type="submit" value="Insert">

</form>
```


=====

category-edit.jsp

```
<form action="<c:url value="/admin/category/update"/>" method="post" enctype="mul

<input type="text" name ="categoryid" value="${cate.categoryid}" hidden="hidden">

<label for="fname">Category name:</label><br>

<input type="text" id="categoryname" name="categoryname" value="${cate.categoryna

<label for="lname">Link images:</label><br>

<input type="text" id="images" name="images" value="${cate.images}"><br>

<c:if test="${cate.images.substring(0,5)=='https'}">

<c:url value="${cate.images }" var="imgUrl"></c:url>

</c:if>

<c:if test="${cate.images.substring(0,5)!='https'}">

<c:url value="/image?fname=${cate.images }" var="imgUrl"></c:url>

</c:if>

<br>

<label for="lname">Upload images:</label><br>

<input type="file" id="images1" name="images1"><br>
```

```
<label for="name">Tên</label><br>

<input type="radio" id="ston" name="status" value="1" ${cate.status==1?'checked':

<label for="css">Hoạt động</label><br>

<input type="radio" id="stoff" name="status" value="0" ${cate.status!=1?'checked'

<label for="javascript">Khóa</label>

<br><br>

<input type="submit" value="Update">

</form>
```

Sửa lần cuối: Thứ Năm, 27 tháng 8 2026, 8:22 AM



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